

Monalisa Dokania

Full-Stack Software Engineer · Founding Engineer Experience

Portfolio: msdokania.github.io · github.com/msdokania · linkedin.com/in/monalisa-dokania

PROFESSIONAL PROFILE

Full-stack engineer focused on building and shipping **products end-to-end**: TypeScript/Python/Go backends, React/Next.js interfaces, real-time event streaming, LLM agent workflows, and production AWS/EKS infrastructure. Founding-engineer experience taking products from architecture through launch. M.S. in Computer Science, Arizona State University (4.0 GPA).

TECHNICAL SKILLS

Backend	Node.js · tRPC · FastAPI · Spring Boot · Microservices · PostgreSQL · Redis · Kafka · SQS
Programming	TypeScript · Python · Go · Java · JavaScript
Cloud & Infra	AWS · EKS · Kubernetes · Terraform · Docker · CI/CD · GitHub Actions · Prometheus · Grafana
AI / Product	LLM applications · RAG · Agents · LangChain · LangGraph · Tools/MCP · Prompt engineering · Vector DBs
Frontend	Next.js · React · TypeScript · Vite · Tailwind · HTML/CSS · Vercel · SSE
Tools	Git · JIRA · Linux · Testing frameworks · Sentry · AI dev tools (Cursor, Claude, Copilot, etc.)

WORK EXPERIENCE

Lead Software Engineer | Jun 2024 – Present

BLUSVN Technologies | Scottsdale, AZ

- Full-stack engineer on a distributed water-delivery platform, owning the **full product lifecycle** from architecture to production across backend, UI design, infrastructure, testing and CI/CD.
- Architected and shipped **100+ backend APIs** (TypeScript/tRPC, PostgreSQL) powering customer, driver, and admin applications; reduced end-to-end latency by **40%** through async work, batching, and indexed query rewrites.
- Engineered high-performance **Golang microservices** with concurrent worker pools and batched notification delivery for order allocation and driver dispatch; validated to **~50K orders/min via load testing**.
- Built **Kafka → Redis event pipeline** for real-time driver tracking, decoupling location ingestion from downstream consumers.
- Built **Python ML inference pipelines** for face/ID validation, **AI-assisted agent workflows** for incident log analysis.
- Designed **React-based customer-facing interfaces** and a **Next.js admin panel** for business admins and managers. Built operational dashboards and features that streamlined daily workflows.
- Designed and operated **production AWS/EKS infrastructure** with Terraform; reduced AWS spend by **12%**, migrated to Karpenter for faster node provisioning, and led unit testing, load and chaos testing to validate SLOs under peak load.
- Partnered with product managers and stakeholders on **technical architecture** and **led engineering delivery & product launch**.
- Managed **CI/CD pipelines** (Jenkins/GitHub Actions); managed Agile sprints, and **incident response** across full delivery cycle.

Graduate Teaching Assistant | May 2023 – May 2024

Arizona State University | Tempe, AZ

- Supported **Software Verification, Validation & Testing** — 900+ students across sections. Led labs and office hours on unit testing, debugging, code coverage, code reviews, and OOP design.

SELECTED PROJECTS

Live Suggestions Copilot — Real-time AI Meeting Assistant (Next.js, TypeScript, Groq, SSE, Tailwind) [GitHub](#) · [Live demo](#)

- Real-time AI copilot that transcribes live meeting audio and generates context-aware suggestions, with a streaming chat panel for follow-ups. Engineered prompt rules for topic locking, recency weighting, anti-repetition, and hallucination control.

Research Assistant — Hybrid RAG + RAGAS Evaluation (Python, LangChain, BM25, Cross-encoders, RAGAS, FastAPI) [GitHub](#)

- Production-grade RAG built in four parts: naive baseline → hybrid retrieval (semantic + BM25) → cross-encoder re-ranking and citation enforcement → **RAGAS evaluation against 60 ground-truth Q&A pairs** measuring faithfulness, answer relevancy.

Multi-Agent LLM Pipeline for Document & Log Analysis (Python, LangChain, FastAPI, Qdrant, Redis RQ) [GitHub](#)

- Five-stage agent system (Planner → Retriever → Analyst → Critic → Reporter) with async PDF processing via Redis RQ, semantic retrieval over Qdrant, and a Critic agent that catches ~30% of unsupported claims before the final report.

EDUCATION

M.S., Computer Science — Arizona State University, Tempe AZ · GPA 4.0 / 4.0 · Aug 2022 – May 2024

B.Tech., Information Technology — MNNIT Allahabad, India · 2011 – 2015